Computer Store

System Requirement List

1. Purpose of the System

The Computer Store Website System is an online platform that allows customers to browse, compare, and purchase computer products conveniently from any location.
It manages products, online sales, promotions, orders, customer accounts, and reviews.
The goal is to provide a smooth, secure, and user-friendly online shopping experience with real-time product information and order tracking.

2. Functional Requirements

A. Product Management
- Display product catalog: computers, laptops, monitors, peripherals, and accessories.
- Product details: name, brand, model, specifications, price, stock, images, category, release date.
- Allow users to search, filter, and sort products.
- Product comparison (side-by-side view of features and prices).
- Show related products and recommendations.

B. Customer Management
- Online registration and login.
- Manage customer profiles: name, contact info, shipping address, order history.
- Allow customers to update their information.

C. Shopping & Checkout
- Add to cart / remove from cart.
- Apply promotions or discount codes.
- Checkout process: shipping details → payment → order confirmation.
- Payment methods: credit/debit card, e-wallet, or online banking.
- Generate online invoice or receipt.
- Order tracking and history page.

D. Product Reviews & Ratings
- Customers can rate and review purchased products.
- Display average ratings and customer feedback on each product page.

E. Promotion & Discount Management
- Admin can create and manage promotions or discount codes.
- Automatically apply valid discounts at checkout.
- Display active promotions on the homepage and product list.

F. Notifications & Alerts
- Send order confirmation and shipping updates via email or SMS.
- Notify customers about promotions, price drops, or restocked products.

G. Reporting & Analytics
- Admin dashboard for viewing total sales, top-selling products, and customer activities.
- Reports for daily/weekly/monthly performance and website traffic.

H. Security & Access Control
- Secure login (HTTPS, encrypted passwords).
- Role-based access: admin, staff, customer.
- Data protection for customer and payment information.

3. Non-Functional Requirements

Reliability: 24/7 website uptime.
Performance: Fast page loading, efficient search, and checkout.
Usability: Modern, intuitive web interface.
Scalability: Can handle many users and products.
Maintainability: Easy to update and manage.
Backup & Recovery: Automatic daily backups.

4. System Components / Modules

1. Product Module
2. Customer Module
3. Booking Module
4. Promotion Module
5. Sales Module
6. Cancellation Module
7. Reporting Module
8. Notification Module
9. Security Module

5. Process Flow Overview

Customer:
Browse Products → Search/Filter → Compare → Add to Cart → Apply Promotions → Checkout → Payment → Receive Confirmation → Review Product

Admin:
Add/Edit Products → Manage Promotions → Monitor Orders → View Reports

6. Data Requirements

Main tables:
- Products
- Customers
- Orders
- Promotions
- Reviews
- Payments

7. System Goals

- Provide a complete online shopping experience.
- Simplify customer purchasing and product comparison.
- Maintain secure and efficient order and payment processing.
- Support promotional and marketing strategies.
- Improve customer satisfaction and retention.

9. Diagrams (Visual Representation)

ERD: Shows relationships between Products, Customers, Orders, Promotions, and Reviews.
DFD: Depicts customer and admin interactions through the website.
Website Flowchart: Shows user journey from browsing to checkout and review.